JEE MAIN ONLINE PAPER 2021

Held on February 24, 2021 (EVENING)

Instructions

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. In the question paper consisting of Physics (Q.no. 1 to 30), Chemistry (Q.no. 31 to 60) and Mathematics (Q.no. 61 to 90). There are two sections for each subject (Section-A : MCQ Type & Section-B : Numerical Response Type). Section-A consists of 20 multiple choice questions & Section-B consists of 10 Numerical Value type Questions. **Candidates have a choice to Answer 5 out of the 10 numerical value answer based questions per section.**
5. There will be only one correct choice in the given four choices in Section-A. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice and zero mark will be awarded for not attempted question. For Section-B questions 4 marks will be awarded for correct answer and zero for unattempted and incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations/written work should be done in the rough sheet provided.

PHYSICS

Section -A

Q.1 When a particle executes SHM, the nature of graphical representation of velocity as a function of displacement is :

(1) circular (2) elliptical

(3) parabolic (4) straight line

Q.2 Two electrons each are fixed at a distance '2d'. A third charge proton placed at the midpoint is displaced slightly by a distance x (x << d) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency : (m = mass of charged particle)

(1) $\left(\frac{2q^2}{\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$ (2) $\left(\frac{\pi\varepsilon_0 md^3}{2q^2}\right)^{\frac{1}{2}}$

(3) $\left(\frac{q^2}{2\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$ (4) $\left(\frac{2\pi\varepsilon_0 md^3}{q^2}\right)^{\frac{1}{2}}$

Q.3 On the basis of kinetic theory of gases, the gas exerts pressure because its molecules :

(1) continuously lose their energy till it reaches wall.

(2) are attracted by the walls of container.

(3) continuously stick to the walls of container.

(4) suffer change in momentum when impinge on the walls of container.

Q.4 A soft ferromagnetic material is placed in an external magnetic field. The magnetic domains

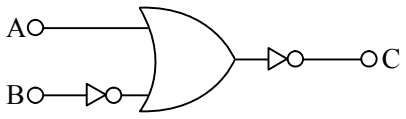
(1) increase in size but no change in orientation.

(2) have no relation with external magnetic field.

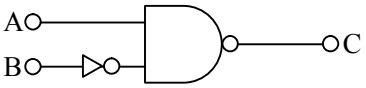
(3) decrease in size and changes orientation.

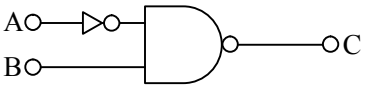
(4) may increase or decrease in size and change its orientation.

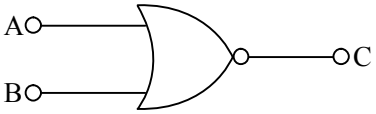
Q.5

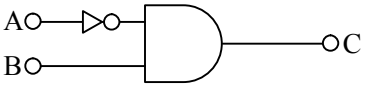


The logic circuit shown above is equivalent to :

(1) 

(2) 

(3) 

(4) 

Q.6 The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of 'L' is 1.0 m from meter scale having a minimum division of 1 mm and time of one complete oscillation is 1.95 s measured from stopwatch of 0.01 s resolution. The percentage error in the determination of 'g' will be :

(1) 1.13% (2) 1.03%

(3) 1.33% (4) 1.30%

Q.7 Given below are two statements :

Statement I : PN junction diodes can be used to function as transistor, simply by connecting two diodes, back to back, which acts as the base terminal.

Statement II : In the study of transistor, the amplification factor β indicates ratio of the collector current to the base current. In the light of the above statements, choose the correct answer from the options given below :

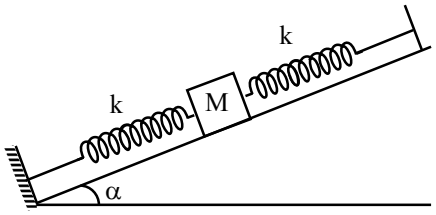
(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is true but Statement II is false

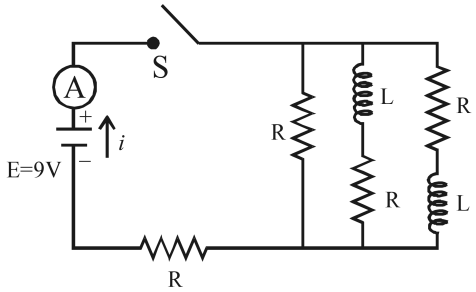
Q.8 In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k, the frequency of oscillation of given body is :



(1) $\frac{1}{2\pi}\sqrt{\frac{k}{2M}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{2k}{Mg\sin\alpha}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{k}{Mg\sin\alpha}}$

Q.9 Figure shows a circuit that contains four identical resistors with resistance R = 2.0 Ω, two identical inductors with inductance L = 2.0 mH and an ideal battery with *emf* E = 9 V. The current 'i' just after the switch 'S' is closed will be



(1) 2.25 A (2) 3.0 A

(3) 3.37 A (4) 9 A

Q.10 The de Broglie wavelength of a proton and α-particle are equal. The ratio of their velocities is :

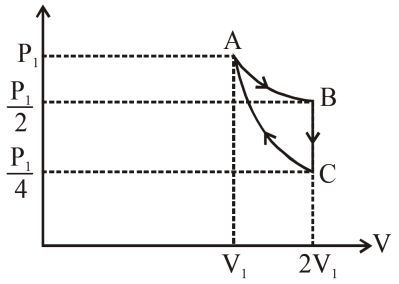
(1) 4 : 3 (2) 4 : 1

(3) 4 : 2 (4) 1 : 4

Q.11 If one mole of an ideal gas at (P_1, V_1) is allowed to expand reversibly and isothermally (A to B) its pressure is reduced to one-half of the original pressure (see figure).

This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value (B $\rightarrow$ C). Then it is restored to its initial state by a reversible adiabatic compression

(C to A). The net workdone by the gas is equal to :



(1) $RT\left(\ln 2 - \frac{1}{(\gamma-1)}\right)$ (2) $-\frac{RT}{2(\gamma-1)}$

(3) 0 (4) RT ln 2

Q.12 An X-ray tube is operated at 1.24 million volt. The shortest wavelength of the produced photon will be :

(1) 10^{-3} nm (2) 10^{-1} nm

(3) 10^{-2} nm (4) 10^{-4} nm

Q.13 Which of the following equations represents a travelling wave ?

(1) $y = A\sin(15x - 2t)$

(2) $y = Ae^{-x^2}(vt + \theta)$

(3) $y = Ae^{x}\cos(\omega t - \theta)$

(4) $y = A\sin x \cos\omega t$

Q.14 According to Bohr atom model, in which of the following transitions will the frequency be maximum ?

(1) n = 4 to n = 3 (2) n = 2 to n = 1

(3) n = 5 to n = 4 (4) n = 3 to n = 2

Q.15 If the source of light used in a Young's double slit experiment is changed from red to violet :

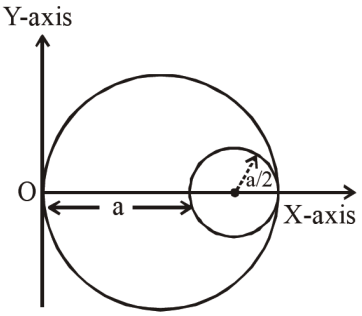
(1) consecutive fringe lines will come closer.

(2) the central bright fringe will become a dark fringe.

(3) the fringes will become brighter.

(4) the intensity of minima will increase.

Q.16 A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius 'a' as shown in figure. The centroid of the remaining circular portion with respect to point 'O' will be :



(1) $\frac{1}{6}a$ (2) $\frac{10}{11}a$

(3) $\frac{5}{6}a$ (4) $\frac{2}{3}a$

Q.17 Zener breakdown occurs in a *p-n* junction having p and n both :

(1) lightly doped and have wide depletion layer.

(2) heavily doped and have narrow depletion layer.

(3) lightly doped and have narrow depletion layer.

(4) heavily doped and have wide depletion layer.

Q.18 Match List-I with List-II

	List-I		List-II
(a)	Source of microwave frequency	(i)	Radioactive decay on nucleus
(b)	Source of infrared frequency	(ii)	Magnetron
(c)	Source of Gamma Rays	(iii)	Inner shell electrons
(d)	Source of Gamma Rays	(iv)	Vibration of atoms and molecules
		(v)	LASER
		(vi)	RC circuit

Choose the correct answer from the options given below :

(1) (a)-(vi), (b)-(iv), (c)-(i), (d)-(v)

(2) (a)-(vi), (b)-(v), (c)-(i), (d)-(iv)

(3) (a)-(ii), (b)-(iv), (c)-(vi), (d)-(iii)

(4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

Q.19 A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin i.e., $ma = -\alpha x^2$. The distance at which the particle stops :

(1) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{2}}$ (2) $\left(\frac{2v_0}{3\alpha}\right)^{\frac{1}{3}}$

(3) $\left(\frac{2v_0^2}{3\alpha}\right)^{\frac{1}{2}}$ (4) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{3}}$

Q.20 A body weighs 49 N on a spring balance at the north pole. What will be its weight recorded on the same weighing machine, if it is shifted to the equator ? (Use $g = \frac{GM}{R^2} = 9.8\ ms^{-2}$ and radius of earth, R = 6400 km.]

(1) 49 N (2) 48.83 N

(3) 49.83 N (4) 49.17 N

Section -B

Q.21 A uniform metallic wire is elongated by 0.04 m when subjected to a linear force F. The elongation, if its length and diameter is doubled and subjected to the same force will be _____ cm.

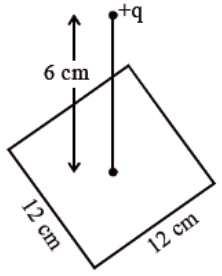
Q.22 A cylindrical wire of radius 0.5 mm and conductivity 5×10^7 S/m is subjected to an electric field of 10 mV/m. The expected value of current in the wire will be $x^3\pi$ mA. The value of x is ____.

Q.23 A uniform thin bar of mass 6 kg and length 2.4 meter is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon is ____ $\times 10^{-1}$ kg m^2.

Q.24 Two solids A and B of mass 1 kg and 2 kg respectively are moving with equal linear momentum. The ratio of their kinetic energies $(K.E.)_A : (K.E.)_B$ will be $\frac{A}{1}$, so the value of will be ___.

Q.25 The root mean square speed of molecules of a given mass of a gas at 27°C and 1 atmosphere pressure is 200 ms^{-1}. The root mean square speed of molecules of the gas at 127°C and 2 atmosphere pressure is $\frac{x}{\sqrt{3}} ms^{-1}$. The value of x will be _____.

Q.26 A point charge of +12 μC is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be _____ $\times 10^3$ Nm^2/C.



Q.27 A signal of 0.1 kW is transmitted in a cable. The attenuation of cable is –5 dB per km and cable length is 20 km. The power received at receiver is 10^{-x} W. The value of x is _____.

[Gain in $dB = 10\log_{10}\left(\frac{P_0}{P_1}\right)$]

Q.28 A series LCR circuit is designed to resonate at an angular frequency $\omega_0 = 10^5$ rad/s. The circuit draws 16 W power from 120 V source at resonance. The value of resistance 'R' in the circuit is ___W.

Q.29 Two cars are approaching each other at an equal speed of 7.2 km/hr. When they see each other, both blow horns having frequency of 676 Hz. The beat frequency heard by each driver will be _____ Hz. [Velocity of sound in air is 340 m/s.]

Q.30 An electromagnetic wave of frequency 3 GHz enters a dielectric medium of relative electric permittivity 2.25 from vacuum. The wavelength of this wave in that medium will be ____ $\times 10^{-2}$ cm.

CHEMISTRY

Section -A

Q.31 What is the correct sequence of reagents used for converting nitrobenzene into m-dibromobenzene ?

(1) $\xrightarrow{NaNO_2}/\xrightarrow{HCl}/\xrightarrow{KBr}/\xrightarrow{H^+}$

(2) $\xrightarrow{Br_2/Fe}/\xrightarrow{Sn/HCl}/\xrightarrow{NaNO_2/HCl}/\xrightarrow{CuBr/HBr}$

(3) $\xrightarrow{Sn/HCl}/\xrightarrow{KBr}/\xrightarrow{Br_2}/\xrightarrow{H^+}$

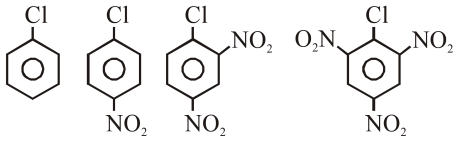
(4) $\xrightarrow{Sn/HCl}/\xrightarrow{Br_2}/\xrightarrow{NaNO_2}/\xrightarrow{NaBr}$

Q.32 Most suitable salt which can be used for efficient clotting of blood will be :-

(1) $NaHCO_3$ (2) $FeSO_4$

(3) $Mg(HCO_3)_2$ (4) $FeCl_3$

Q.33 The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is-



(1) (iv) < (iii) < (ii) < (i)

(2) (iv) < (i) < (ii) < (iii)

(3) (iv) < (i) < (iii) < (ii)

(4) (i) < (ii) < (iii) < (iv)

Q.34 According to Bohr's atomic theory :-

(A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$

(B) The product of velocity (v) of electron and principal quantum number (n), 'vn' $\propto Z^2$.

(C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$

(D) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$

Choose the most appropriate answer from the options given below :

(1) (C) Only

(2) (A) Only

(3) (A), (C) and (D) only

(4) (A) and (D) only

Q.35 Match list - I and List - II.

List-I	List-II
(a) $R-\overset{\overset{O}{\|}}{C}-Cl \rightarrow R-CHO$	(i) Br_2 /NaOH
(b) $R-CH_2-COOH \rightarrow R-\underset{\underset{Cl}{\|}}{CH}-COOH$	(ii) H_2 / Pd–$BaSO_4$
(c) $R-\overset{\overset{O}{\|}}{C}-NH_2 \rightarrow R-NH_2$	(iii) Zn(Hg)/Conc.HCl
(d) $R-\overset{\overset{O}{\|}}{C}-CH_3 \rightarrow R-CH_2-CH_3$	(iv) Cl_2 /Red P, H_2O

Choose the correct answer from the options given below :

(1) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)

(2) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

(3) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)

(4) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

Q.36 The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MNO_4^{3-} respectively are :

(1) 5.82, 0 and 0 BM

(2) 4.90, 0 and 1.73 BM

(3) 5.92, 4.90 and 0 BM

(4) 4.90, 0 and 2.83 BM

Q.37 Match List-I with List-II :

	List-I (Salt)		List-II (Flame colour wavelength)
(a)	LiCl	(i)	455.5 nm
(b)	NaCl	(ii)	670.8 nm
(c)	RbCl	(iii)	780.0 nm
(d)	CsCl	(iv)	589.2 nm

Choose the correct answer from the options given below :

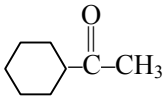
(1) (a)–(iv), (b)–(ii), (c)–(iii), (d)–(i)

(2) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)

(3) (a)–(i), (b)–(iv), (c)–(ii), (d)–(iii)

(4) (a)–(ii), (b)–(iv), (c)–(iii), (d)–(i)

Q.38 Which one of the following carbonyl compounds cannot be prepared by addition of water on an alkyne in the presence of $HgSO_4$ and H_2SO_4 ?

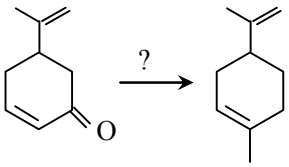
(1) $CH_3-\overset{\overset{O}{\|}}{C}-H$ (2) 

(3) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H$ (4) $CH_3-\overset{\overset{O}{\|}}{C}-CH_2CH_3$

Q.39 In polymer Buna-S: 'S' stands for :-

(1) Sulphonation (2) Strength

(3) Sulphur (4) Styrene

Q.40 

Which of the following reagent is suitable for the preparation of the product in the above reaction ?

(1) $NaBH_4$

(2) $NH_2-NH_2/\ C_2H_5\overset{\ominus}{O}\overset{\oplus}{N}a$

(3) Ni/H_2

(4) Red P + Cl_2

Q.41 Match List-I and List-II.

	List-I		List-II
(a)	Valium	(i)	Antifertility drug
(b)	Morphine	(ii)	Pernicious anaemia
(c)	Norethindrone	(iii)	Analgesic
(d)	Vitamin B_{12}	(iv)	Tranquilizer

(1) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)

(2) (a)–(iv), (b)–(iii), (c)–(i), (d)–(ii)

(3) (a)–(ii), (b)–(iv), (c)–(iii), (d)–(i)

(4) (a)–(i), (b)–(iii), (c)–(iv), (d)–(ii)

Q.42 Match List-I with List-II.

	List-I (Metal)		**List-II** (Ores)
(a)	Aluminium	(i)	Siderite
(b)	Iron	(ii)	Calamine
(c)	Copper	(iii)	Kaolinite
(d)	Zinc	(iv)	Malachite

Choose the correct answer from the options given below :

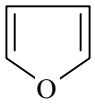
(1) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)

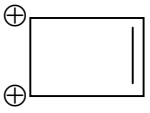
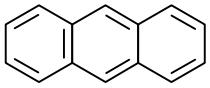
(2) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)

(3) (a)–(i), (b)–(ii), (c)–(iii), (d)–(iv)

(4) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

Q.43 Which one of the following compounds is nonaromatic ?

(1)  (2) 

(3)  (4) 

Q.44 What is the correct order of the following elements with respect to their density ?

(1) Cr < Zn < Co < Cu < Fe

(2) Zn < Cu < Co < Fe < Cr

(3) Zn < Cr < Fe < Co < Cu

(4) Cr < Fe < Co < Cu < Zn

Q.45 Given below are two statements :-

Statement I : The value of the parameter "Biochemical Oxygen Demand (BOD)" is important for survival of aquatic life.

Statement II : The optimum value of BOD is 6.5 ppm. In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

Q.46 The incorrect statement among the following is

(1) $VOSO_4$ is a reducing agent

(2) Cr_2O_3 is an amphoteric oxide

(3) RuO_4 is an oxidizing agent

(4) Red colour of ruby is due to the presence of Co^{3+}

Q.47 The correct shape and I–I–I bond angles respectively in I_3^- ion are :-

(1) Distorted trigonal planar; 135° and 90°

(2) T-shaped; 180° and 90°

(3) Trigonal planar; 120°

(4) Linear; 180°

Q.48 Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Hydrogen is the most abundant element in the Universe, but it is not the most abundant gas in the troposphere.

Reason R : Hydrogen is the lightest element. In the light of the above statements, choose the correct answer from the options given below :

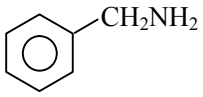
(1) A is true but R is false

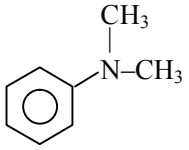
(2) Both A and R are true and R is the correct explanation of A

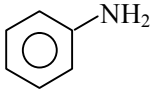
(3) A is false but R is true

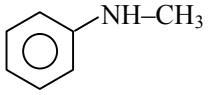
(4) Both A and R are true but R is NOT the correct explanation of A

Q.49 The diazonium salt of which of the following compounds will form a coloured dye on reaction with b-Naphthol in NaOH ?

(1) 

(2) 

(3) 

(4) 

Q.50 The correct set from the following in which both pairs are in correct order of melting point is :

(1) LiF > LiCl ; MgO > NaCl

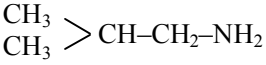
(2) LiCl > LiF ; NaCl > MgO

(3) LiF > LiCl ; NaCl > MgO

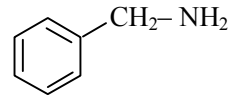
(4) LiCl > LiF ; MgO > NaCl

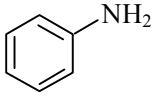
Section -B

Q.51 The total number of amines among the following which can be synthesized by Gabriel synthesis is _____.

(1) 

(2) $CH_3CH_2NH_2$

(3) 

(4) 

Q.52 Among the following allotropic forms of sulphur, the number of allotropic forms, which will show paramagnetism is _____.

(A) α-sulphur (B) β-sulphur

(C) S_2-form (4) None of these

Q.53 The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y. The value of y is _____.

Q.54 The volume occupied by 4.75 g of acetylene gas at 50°C and 740 mmHg pressure is _____ L. (Rounded off to the nearest integer) [Given R = 0.0826 L atm K^{-1} mol^{-1}]

Q.55 C_6H_6 freezes at 5.5°C. The temperature at which a solution 10 g of C_4H_{10} in 200 g of C_6H_6 freeze is _____ °C. (The molal freezing point depression constant of C_6H_6 is 5.12°C/m.)

Q.56 The magnitude of the change in oxidising power of the MnO_4^- / Mn^{2+} couple is x × 10^{-4} V, if the H^+ concentration is decreased from 1 M to 10^{-4} M at 25°C. (Assume concentration of MnO_4^- and Mn^{2+} to be same on change in H+ concentration). The value of x is _____. (Rounded off to the nearest integer) [Given : $\frac{2.303RT}{F} = 0.059$]

Q.57 The solubility product of PbI_2 is 8.0 × 10^{-9}. The solubility of lead iodide in 0.1 molar solution of lead nitrate is x × 10^{-6} mol/L. The value of x is _____. (Rounded off to the nearest integer) [Given : $\sqrt{2}$ = 1.41]

Q.58 Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25°C. After 9 h, the fraction of sucrose remaining is f. The value of $\log_{10}\left(\frac{1}{f}\right)$ is _____ × 10^{-2}. (Rounded off to the nearest integer) [Assume : ln 10 = 2.303, ln 2 = 0.693]

Q.59 1.86 g of aniline completely reacts to form acetanilide. 10% of the product is lost during purification. Amount of acetanilide obtained after purification (in g) is _____ $\times 10^{-2}$.

Q.60 Assuming ideal behaviour, the magnitude of log K for the following reaction at 25°C is $x \times 10^{-1}$. The value of x is ____. (Integer answer)

$$3HC \equiv CH_{(g)} \rightleftharpoons C_6H_6(\ell)$$

[Given: $\Delta_f G^o(HC \equiv CH) = -2.04 \times 10^5$ J mol^{-1}; $\Delta_f G^o(C_6H_6) = -1.24 \times 10^5$ J mol^{-1}; R = 8.314 J K^{-1} mol^{-1}]

MATHEMATICS

Section -A

Q.61 For the statements p and q, consider the following compound statements :

(a) $(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$

(b) $((p \vee q) \wedge \sim p) \rightarrow q$

Then which of the following statements is correct?

(1) (a) and (b) both are not tautologies.

(2) (a) and (b) both are tautologies.

(3) (a) is a tautology but not (b).

(4) (b) is a tautology but not (a).

Q.62 Let a, b $\in$ R. If the mirror image of the point P(a, 6, 9) with respect to the line $\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is (20, b, –a –9), then |a + b| is equal to :

(1) 88 (2) 86

(3) 84 (4) 90

Q.63 The vector equation of the plane passing through the intersection of the planes $\vec{r}.(\hat{i}+\hat{j}+\hat{k}) = 1$ and $\vec{r}.(\hat{i}-2\hat{j}) = -2$, and the point (1, 0, 2) is :

(1) $\vec{r}.(\hat{i}+7\hat{j}+3\hat{k}) = \frac{7}{3}$ (2) $\vec{r}.(3\hat{i}+7\hat{j}+3\hat{k}) = 7$

(3) $\vec{r}.(\hat{i}+7\hat{j}+3\hat{k}) = 7$ (4) $\vec{r}.(\hat{i}-7\hat{j}+3\hat{k}) = \frac{7}{3}$

Q.64 If P is a point on the parabola $y = x^2 + 4$ which is closest to the straight line y = 4x – 1, then the co-ordinates of P are :

(1) (3, 13) (2) (1, 5)

(3) (–2, 8) (4) (2, 8)

Q.65 The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 20 seconds at the speed of 432 km/hour, the angle of elevation changes to 30°. If the jet plane is flying at a constant height, then its height is :

(1) $1800\sqrt{3}$ m (2) $3600\sqrt{3}$ m

(3) $2400\sqrt{3}$ m (4) $1200\sqrt{3}$ m

Q.66 If $n \geq 2$ is a positive integer, then the sum of the series $^{n+1}C_2 + 2(^2C_2 + {}^3C_2 + {}^4C_2 + ...+ {}^nC_2)$ is:

(1) $\frac{n(n-1)(2n+1)}{6}$ (2) $\frac{n(n+1)(2n+1)}{6}$

(3) $\frac{n(2n+1)(3n+1)}{6}$ (4) $\frac{n(n+1)^2(n+2)}{12}$

Q.67 Let f : **R** $\rightarrow$ **R** be defined as,

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4 \end{cases}$$

Let A = {x $\in$ R : f is increasing}. Then A is equal to :

(1) $(-\infty, -5) \cup (4, \infty)$ (2) $(-5, \infty)$

(3) $(-\infty, -5) \cup (-4, \infty)$ (4) $(-5, -4) \cup (4, \infty)$

Q.68 Let f be a twice differentiable function defined on R such that f(0) = 1, f '(0) = 2 and f '(x) $\neq$ 0 for all x $\in$ R. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f'(x) \end{vmatrix} = 0$ for all x $\in$ R, then the value of f(1) lies in the interval:

(1) (9, 12) (2) (6, 9)

(3) (0, 3) (4) (3,6)

Q.69 For which of the following curves, the line $x + \sqrt{3}y = 2\sqrt{3}$ is the tangent at the point $\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}\right)$?

(1) $x^2 + y^2 = 7$ (2) $y^2 = \frac{1}{6\sqrt{3}}x$

(3) $2x^2 - 18y^2 = 9$ (4) $x^2 + 9y^2 = 9$

Q.70 The value of the integral, $\int_1^3 [x^2 - 2x - 2]dx$ where [x] denotes the greatest integer less than or equal to x, is :

(1) $-\sqrt{2} - \sqrt{3} + 1$ (2) $-\sqrt{2} - \sqrt{3} - 1$

(3) –5 (4) –4

Q.71 A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is

(1) $\frac{1}{\sqrt{7}}$ (2) $2\sqrt{2}-1$

(3) $\sqrt{7}-1$ (4) $\frac{1}{2\sqrt{2}}$

Q.72 The negative of the statement $\sim p \wedge (p \vee q)$ is

(1) $\sim p \vee q$ (2) $p \vee \sim q$

(3) $\sim p \wedge q$ (4) $p \wedge \sim q$

Q.73 If the curve $y = ax^2 + bx + c$, $x \in R$, passes through the point (1, 2) and the tangent line to this curve at origin is $y = x$, then the possible values of a, b, c are :

(1) $a = \frac{1}{2}, b = \frac{1}{2}, c = 1$

(2) $a = 1, b = 0, c = 1$

(3) $a = 1, b = 1, c = 0$

(4) $a = -1, b = 1, c = 1$

Q.74 The area of the region :

$R = \{(x, y) : 5x^2 \le y \le 2x^2 + 9\}$ is :

(1) $11\sqrt{3}$ square units (2) $12\sqrt{3}$ square units

(3) $9\sqrt{3}$ square units (4) $6\sqrt{3}$ square units

Q.75 If a curve $y = f(x)$ passes through the point (1, 2) and satisfies $x\frac{dy}{dx} + y = bx^4$, then for what value of b, $\int_1^2 f(x)\,dx = \frac{62}{5}$?

(1) 5 (2) 10

(3) $\frac{62}{5}$ (4) $\frac{31}{5}$

Q.76 Let f(x) be a differentiable function defined on [0, 2] such that $f'(x) = f'(2 - x)$ for all $x \in (0, 2)$, $f(0) = 1$ and $f(2) = e^2$. Then the value of $\int_0^2 f(x)dx$ is :

(1) $1 - e^2$ (2) $1 + e^2$

(3) $2(1 - e^2)$ (4) $2(1 + e^2)$

Q.77 Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2B^2 - B^2A^2)X = O$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has :

(1) no solution

(2) exactly two solutions

(3) infinitely many solutions

(4) a unique solution

Q.78 Let a, b, c be in arithmetic progression. Let the centroid of the triangle with vertices (a, c), (2, b) and (a, b) be $\left(\frac{10}{3}, \frac{7}{3}\right)$. If α, β are the roots of the equation $ax^2 + bx + 1 = 0$, then the value of $\alpha^2 + \beta^2 - \alpha\beta$ is :

(1) $\frac{71}{256}$ (2) $\frac{69}{256}$

(3) $-\frac{69}{256}$ (4) $-\frac{71}{256}$

Q.79 For the system of linear equations :

$x - 2y = 1$, $x - y + kz = -2$, $ky + 4z = 6$, $k \in R$,

consider the following statements :

(A) The system has unique solution if $k \ne 2$, $k \ne -2$.

(B) The system has unique solution if $k = -2$.

(C) The system has unique solution if $k = 2$.

(D) The system has no-solution if $k = 2$.

(E) The system has infinite number of solutions if $k \ne -2$.

Which of the following statements are correct ?

(1) (C) and (D) only (2) (B) and (E) only

(3) (A) and (E) only (4) (A) and (D) only

Q.80 The probability that two randomly selected subsets of the set {1, 2, 3, 4, 5} have exactly two elements in their intersection, is :

(1) $\frac{65}{2^7}$ (2) $\frac{65}{2^8}$

(3) $\frac{135}{2^9}$ (4) $\frac{35}{2^7}$

Section–B

Q.81 For integers n and r,

let $\binom{n}{r} = \begin{cases} {}^{n}C_{r}, & \text{if } n \geq r \geq 0 \\ 0, & \text{otherwise} \end{cases}$

The maximum value of k for which the sum $\sum_{i=0}^{k}\binom{10}{i}\binom{15}{k-i}+\sum_{i=0}^{k+1}\binom{12}{i}\binom{13}{k+1-i}$ exists, is equal to _____.

Q.82 Let λ be an integer. If the shortest distance between the lines $x - \lambda = 2y - 1 = -2z$ and $x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is _____.

Q.83 If $a + \alpha = 1$, $b + \beta = 2$ and $af(x)\,\alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}$, $x \neq 0$, then the value of expression $\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}$ is _____.

Q.84 Let a point P be such that its distance from the point (5, 0) is thrice the distance of P from the point (– 5, 0). If the locus of the point P is a circle of radius r, then $4r^2$ is equal to _____.

Q.85 If the area of the triangle formed by the positive x-axis, the normal and the tangent to the circle $(x - 2)^2 + (y - 3)^2 = 25$ at the point (5, 7) is A, then 24A is equal to_____.

Q.86 If the variance of 10 natural numbers 1, 1, 1, ..., 1, k is less than 10, then the maximum possible value of k is _____.

Q.87 The sum of first four terms of a geometric progression (G.P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is a, then 2α is _____.

Q.88 The students S_1, S_2, ... , S_{10} are to be divided into 3 groups A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is _____.

Q.89 Let $i = \sqrt{-1}$. If $\frac{(-1+i\sqrt{3})^{21}}{(1-i)^{24}}+\frac{(1+i\sqrt{3})^{21}}{(1+i)24} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5}(j+5)^2 - \sum_{j=0}^{n+5}(j+5)$ is equal to _____.

Q.90 The number of the real roots of the equation $(x + 1)^2 + |x - 5| = \frac{27}{4}$ is _____.

JEE MAIN ONLINE PAPER 2021

Held on February 24, 2021 (Evening)

Hints & Solutions

PHYSICS

1.[2] For a particle executing SHM,

$x = A\sin(\omega t + \phi)$

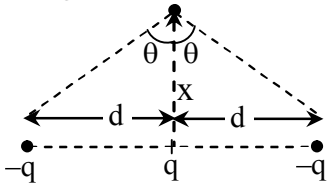
$v = \omega A\cos(\omega t + \phi)$

$\Rightarrow \frac{v^2}{\omega^2 A^2} + \frac{x^2}{A^2} = 1 \Rightarrow$ equation of ellipse

between v and x

Hence option (2)

2.[3] From the given condition, we have



$F_{netq} = -[2F_{q/q}\cos\theta]$

$$F_{netq} = -2\cdot\frac{1}{4\pi\varepsilon_0}\cdot\frac{q^2}{(\sqrt{d^2+x^2})^2}\cdot\frac{x}{\sqrt{d^2+x^2}}$$

$$= \frac{q^2}{2\pi\varepsilon_0}\frac{x}{(d^2+x^2)^{3/2}}$$

For $x << d$,

$$F_{net}\, q = -\frac{q^2}{2\pi\varepsilon_0 d^3}x$$

$$\therefore\ a = \frac{q^2}{2\pi\varepsilon_0\cdot md^3}x$$

Comparing with equation of SHM ($a = -\omega^2 x$)

$$\therefore\ \omega = \sqrt{\frac{q^2}{2\pi\varepsilon_0 md^3}}$$

Hence option (3) is correct

3.[4] From the assumption of KTG, the molecules of gas collide with the walls and suffers momentum change which results in force on the wall and hence pressure.

Hence option (4) is correct

4.[4] Soft ferromagnetic materials are materials which can be easily magnetised and demagnetised by external magnetic field. When external field is applied, the domains experiences a net torque hence change its orientation.

Hence option (4) is correct

5.[4] Truth table of the given gate :

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Truth table of option (1)

A	B	C
0	0	1
0	1	1
1	0	0
1	1	1

Truth table of option (2)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	1

Truth table of option (3)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Truth table of option (4)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Since option (1) has same truth table, hence answer is option (4) is correct.

6.[1] $T = 2\pi\sqrt{\frac{\ell}{g}}$

$$g = \frac{4\pi^2\ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta\ell}{\ell} + \frac{2\Delta T}{T}$$

$$\frac{\Delta g}{g} = \frac{1\times10^{-3}}{1} + 2\times\frac{0.01}{1.95}$$

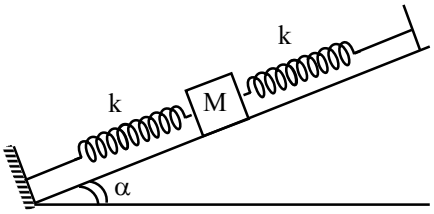
$\frac{\Delta g}{g} = 0.0113$ or 1.13%

option (1) is correct

7.[1] Back to back diode will not the make a transistor

$\beta = \frac{i_c}{i_b}$

8.[3]



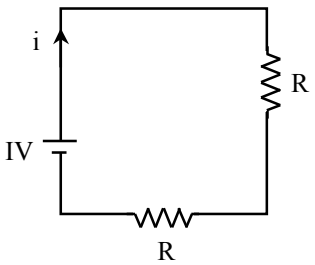
$K_{eq} = K_1 + K_2 = K + K = 2K$

$T = 2\pi\sqrt{\frac{m}{K_{eq}}} = 2\pi\sqrt{\frac{m}{2K}}$

$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{2K}{m}}$

(Option 3) is correct

9.[1] Just after the switch is closed, inductor will behave like infinite resistance (open circuit) so the circuit will look like



$i = \frac{9}{R+R} = \frac{9}{4} = 2.25$

Option (1) is correct.

10.[2] $\lambda = \frac{h}{mv}$

$\lambda_p = \lambda_\alpha$

$m_P v_P = m_\alpha v_\alpha$

$m_P v_P = 4m_p v_\alpha$ $(m_\alpha = 4m_P)$

$\frac{V_P}{V_\alpha} = 4$

(Option 2) is correct

11.[1] A – B = isothermal process

$W_{AB} = P_1V_1 \ln\left[\frac{2V_1}{V_1}\right] = P_1V_1 \ln(2)$

B – C $\rightarrow$ Isochoric process

$W_{BC} = 0$

C – A $\rightarrow$ Adiabatic process

$W_{CA} = \frac{P_1V_1 - \frac{P_1}{4}\times 2V_1}{1-\gamma} = \frac{P_1V_1\left[1-\frac{1}{2}\right]}{1-\gamma} = \frac{P_1V_1}{2(1-\gamma)}$

$W_{net} = W_{AB} + W_{BC} + W_{CA}$ $\{P_1V_1 = RT\}$

$= P_1V_1\ln(2) + 0 + \frac{P_1V_1}{2(1-\gamma)}$

Option (1) is correct.

12.[1] $\lambda_{min} = \frac{1240}{\Delta V}$(nm)

$= \frac{1240}{1.24\times 10^6} = 10^{-3}$ nm

Option (1) is correct.

13.[1] y = F(x, t)

For travelling wave y should be linear function of x and t and they must exist as (x ± vt)

y = Asin(15x – 2t) $\rightarrow$ linear function in x and t

Option (1) is correct.

14.[2] $\Delta E = 13.6\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = h\nu$

It is maximum if $n_1 = 1$ and $n_2 = 2$

n = 5 ..... –0.544 eV

n = 4 ..... –0.850 eV

n = 3 ..... –1.511 eV

n = 2 ..... –3.4 eV

n = 1 ..... –13.6 eV

Option (2) is correct.

15.[1] $\beta = \frac{\lambda D}{d}$

$\lambda_R > \lambda_V$

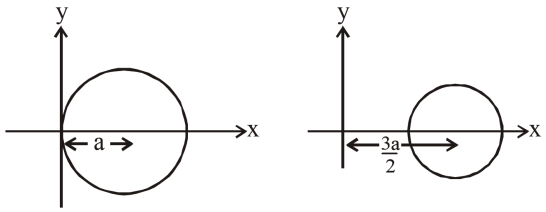
$\beta_R = \frac{\lambda_R D}{d}$ and $\beta_V = \frac{\lambda_v D}{d}$

$\beta_R > \beta_V$

Fringe pattern will shrink.

Option (1) is correct.

16.[3]



Let σ be the uniform mass density of disc then

$$X_{COM} = \frac{(\sigma\pi a^2)a - \sigma\pi\left(\frac{a^2}{4}\right)\times\frac{3a}{2}}{\sigma\pi a^2 - \frac{\sigma\pi a^2}{4}}$$

$$= \frac{a - \frac{3a}{8}}{1 - \frac{1}{4}} = \frac{5a}{6}$$

Option (2) is correct.

17.[2] Zener diode is heavily doped and have narrow depletion layer.
Option (2) is correct.

18.[4] (a) Source of microwave frequency is magnetron.
(b) Source of infrared frequency is vibration of atoms and molecules.
(c) Source of Gamma rays is radioactive decay of nucleus
(d) Source of X-rays inner shell electron transition.
Option (4) is correct.

19.[4] $F = -\alpha x^2$

$ma = -\alpha x^2$

$$a = \frac{-\alpha x^2}{m}$$

$$\frac{vdv}{dx} = -\frac{\alpha}{m}x^2$$

$$\int_{v_0}^{0} vdv = \int_{0}^{x} -\frac{\alpha}{m}x^2 dx$$

$$\left(\frac{v^2}{2}\right)_{v_0}^{0} = -\frac{\alpha}{m}\left(\frac{x^3}{3}\right)_0^x$$

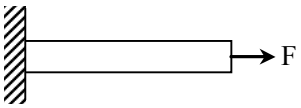
$$\frac{-v_0^2}{2} = -\frac{\alpha}{m}\frac{x^3}{3}$$

$$\boxed{x = \left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{3}}}$$

Option(4) is most suitable option as (m) is not given in any option

20.[2] Weight of pole = mg = 49 N
At equator due to rotation = $g_e = g - R\omega^2$
so $W = mg_e = m(g - R\omega^2)$
$\therefore$ $W_P > W_e$ $\quad W_P = 49$ N
So, $W_e = 48.83$ N. $\quad W_e < 49$ N
Option (2) is correct.

21.[2]



$$F = Y.A.\frac{\Delta\ell}{\ell}$$

$$\Delta\ell = \frac{F}{Y.A.}\cdot\ell$$

$$\Delta\ell = \frac{F\cdot\ell}{Y.\pi r^2}$$

$$\Delta\ell \propto \frac{\ell}{r^2}$$

$$\frac{\Delta\ell_2}{\Delta\ell_1} = \left(\frac{\ell_2}{\ell_1}\right)\left(\frac{r_1}{r_2}\right)^2$$

$$= (2)\left(\frac{1}{2}\right)^2$$

$$\frac{\Delta\ell_2}{\Delta\ell_1} = \frac{1}{2}$$

$$\Delta\ell_2 = \frac{\Delta\ell_1}{2}$$

$$= \frac{0.04}{2}$$

$= 0.02$ m

$\Delta\ell_2 = 2$cm

Ans.=2

22.[5] Conductivity $s = 5 \times 10^7$ S/m
Radius r = 0.5 mm = 5×10^{-4} m

$$E = 10\times10^{-3}\frac{V}{m}$$

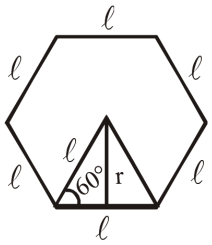
$J = \sigma E = 10\ ´10^{-3} \times 5 \times 10^7$

$J = 5 \times 10^5$

$$\frac{i}{A} = 5\times10^5$$

$i = 5 \times 10^5 \times \pi r^2$
$= 5 \times 10^5 \times \pi \times (5 \times 10^{-4})^2$
$= 125\pi \times 10^{-3}$ Amp
$i = 125\ \pi$ mA
$\boxed{x = 5}$
Ans. 5

23.[8]



m = mass of one side of hexagon = 1 kg

$6\ell = 2.4,\ \ell = 0.4$ m

$\sin 60° = \frac{r}{\ell}$

$r = \ell \sin 60° = \frac{\ell\sqrt{3}}{2}$

MOI, $I = \left[\frac{m\ell^2}{12} + mr^2\right]6$

$\left[\frac{m\ell^2}{12} + m\left(\frac{\ell\sqrt{3}}{2}\right)^2\right]6$

$= 5\ m\ell^2$
$= 5 \times 1 \times 0.16$
$= 0.8$
$I = 8 \times 10^{-1}$ kg m^2 Ans. (2)

24.[2] Kinetic energy $K = \frac{P^2}{2m}$, $(P_A = P_B)$

$K \propto \frac{1}{m}$

$\frac{K_A}{K_B} = \frac{m_B}{m_A} = \frac{2}{1}$
Ans. 2

25.[400] $v_{rms} = \sqrt{\frac{3RT}{M}}$

$v_{rms} \propto \sqrt{T}$

$\frac{(v_{rms})_2}{(v_{rms})_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{400}{300}} = \frac{2}{\sqrt{3}}$

$(v_{rms})_2 = \frac{2}{\sqrt{3}}(v_{rms})_1$

$= \frac{2}{\sqrt{3}} \times 200$

$(v_{rms})_2 = \frac{400}{\sqrt{3}}$ m/s

26.[226] From symmetry $\phi = \frac{1}{6}\left(\frac{q}{\varepsilon_0}\right)$

$= \frac{12 \times 10^{-6}}{6 \times 8.85 \times 10^{-12}}$

$= 225.98 \times 10^3\ \frac{Nm^2}{s}$

$\simeq 226 \times 10^3\ \frac{Nm^2}{C}$

27.[8] Sound level decreases by 5dB every km so sound level decreased in 20 km = 100 dB

$\beta_2 - \beta_1 = 10 \log_{10} \frac{I_2}{I_1}$

$-100 = 10\log_{10} \frac{I_2}{I_1} \Rightarrow \frac{I_1}{I_2} = 10^{10}$

$I_2 = 10^{-10}\ I_1 \Rightarrow P_2 = 10^{-10}\ P_1 = 10^{-8}$ W
$x = 8$ Ans. 8

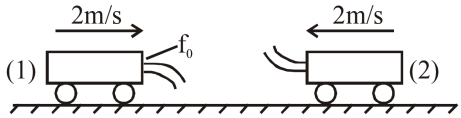
28.[900] At resonance

$P = \frac{V^2}{R}$

$R = \frac{V^2}{P} = \frac{(120)^2}{16}$

$= 900\Omega$

29.[8]



Frequency of sound heard by car-1, which comes by reflection from car-2

$f_1 = f_0\left(\frac{340+2}{340-2}\right)\left(\frac{340+2}{340-2}\right) = f_0\left(\frac{342}{338}\right)^2$

Frequency of sound coming directly from car-2

$f_2 = f_0\left(\frac{340+2}{340-2}\right)$

$= f_1 - f_2 = f_0\left(\frac{342}{338}\right)\left(\frac{342}{338} - 1\right) = 8.09 \simeq 8$

30.[667] ℓ in vaccum $= \frac{c}{f} = \frac{3\times10^8}{3\times10^9} = 0.1m$

$\therefore$ λ in medium $= \frac{0.1}{\mu}$

Where refractive index

$\mu = \sqrt{\mu_r\varepsilon_r}$

Assuming non-magnetic material $\mu_r = 1$

$\therefore$ $\mu = \sqrt{2.25} = 1.5$

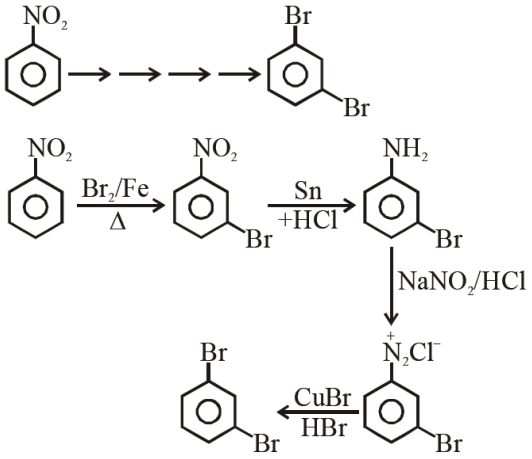
$\lambda_m = \frac{0.1}{1.5} = \frac{1}{15} m = 6.67cm$

$= 667 \times 10^{-2}$ cm

Ans. 667

CHEMISTRY

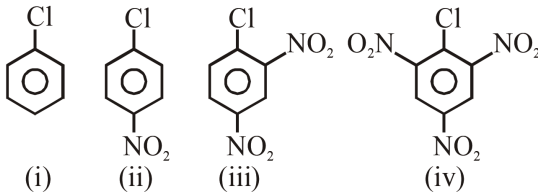
31.[2] Correct sequence of reagents for the following conversion



32.[4] Blood : negatively charged sol According to Hardly-schulz rule, for the negatively charged sol, most (+) ve ion is needed for its efficient coagulation.

Ans. : $FeCl_3$

33.[4] For nucleophile substitution in aromatic halides



Correct order is :

(i) < (ii) < (iii) < (iv)

More No. of NO_2 substituted aromatic halide, increase the rate of nucleophile substitution reaction in aromatic halides.

34.[4] **Official Answer by NTA (3)**

According to Bohr's theory

(A) KE $= 13.6\frac{Z^2}{n^2}\frac{eV}{atom} \Rightarrow KE\alpha\frac{z^2}{n^2}$

(B) speed of $e^-\alpha\frac{z}{n}$

$\therefore$ $\boxed{v\times n\alpha z}$

(C) Frequency of revolution of $e^- = \frac{v}{2\pi r}$

$\therefore$ $\boxed{\text{frequency } \alpha\frac{z^2}{n^3}}$

(D) $F = \frac{\kappa q_1 q_2}{r^2} = \frac{kze^2}{r^2}$ $\left\{r\alpha\frac{n^2}{z}\right.$

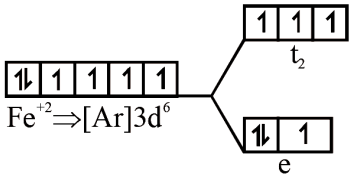
$\Rightarrow F\alpha\frac{z}{\left(\frac{n^2}{z}\right)^2}$

$\Rightarrow \boxed{F\alpha\frac{z^3}{n^4}}$

35.[3]

(a)	$R-\overset{O}{\overset{\|}{C}}-Cl \xrightarrow[Pd\text{-}BaSO_4]{H_2} R-\overset{O}{\overset{\|}{C}}-H$ Rosenmund Reduction	(a) – (ii)
(b)	$R-CH_2-COOH \xrightarrow[H_2O]{Cl_2/P} R-\underset{Cl}{CH}-COOH$ HVZ reaction	(b) – (iv)
(c)	$R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[NaOH]{Br_2} R-NH_2$ Hoffmann Bromamide reaction	(c) – (i)
(d)	$R-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow[con.\ HCl]{Zn(Hg)} R-CH_2-CH_3$ Clemmenson reduction	(d) – (iii)

36.[2] (i) $[FeCl_4]^{-2}$

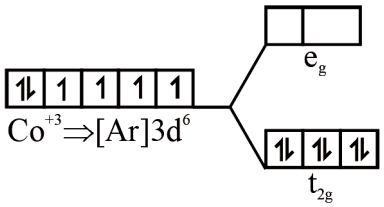


$\mu = \sqrt{n(n+2)}BM$

$= \sqrt{4(4+2)}BM$

$= \sqrt{24}BM \Rightarrow 4.90\,BM$

(ii) $[Co(C_2O_4)_3]^{-3}$



$\mu = 0$

(iii) MnO_4^{-2}

$Mn^{+6} \Rightarrow [Ar]\ 3d^1 \quad \mu = \sqrt{n(n+2)}BM$

$= \sqrt{1(1+2)}\ BM$

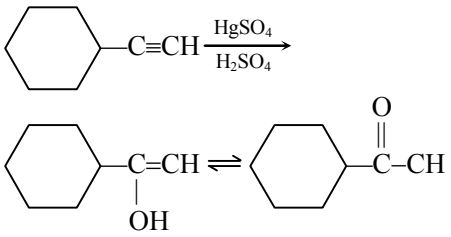
$= \sqrt{3}BM \Rightarrow 1.73\ BM$

37.[4]

	Colour	λ/nm
Li	Crimson red	670.8
Na	Yellow	589.2
Rb	Red violet	780.0
Cs	Blue	455.5

38.[3] Reaction of $HgSO_4$/dil.H_2SO_4 with alkyne gives addition of water as per markonikoff's rule.

(1) $HC{\equiv}CH \xrightarrow[H_2SO_4]{HgSO_4} CH_2{-}CH(OH) \rightleftharpoons CH_3{-}C(=O){-}H$

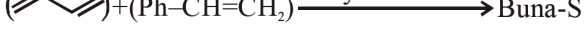
(2) 

(3) $CH_3{-}C{\equiv}CH \xrightarrow[H_2SO_4]{HgSO_4} CH_3{-}C(OH){=}CH_2 \rightleftharpoons CH_3{-}C(=O){-}CH_3$

Hence $CH_3{-}CH_2{-}CHO$ cannot be form.

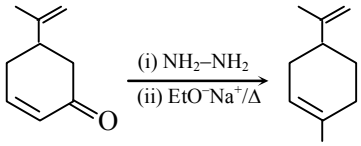
(4) $CH_3{-}C{\equiv}C{-}CH_3 \xrightarrow[H_2SO_4]{HgSO_4} CH_3{-}C(OH){=}CH{-}CH_3 \rightleftharpoons CH_3{-}C(=O){-}CH_2{-}CH_3$

39.[4] BUNA-S, 'S' stand for styrene



Buta -1, 3-diene styrene

40.[2]



To reduce the carbonyl groups into alkane wolf – kischner reduction is used, without affecting the double bond.

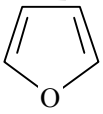
41.[2]

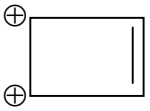
(a) Valium – Tranquilizer	(a)-(iv)
(b) Morphine – Analgesic	(b)-(iii)
(c) Norethindrone – Antifertility Drug	(c)-(i)
(d) Vitamin B12 – Pernicious anaemia	(d)-(ii)

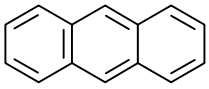
42.[4] Siderite – $FeCO_3$
Calamine – $ZnCO_3$
Kaolinite – $Al_2(OH)_4.Si_2O_5$
Malachite – $Cu(OH)_2.CuCO_3$

43.[1] For the following ion/compounds

(1)  –Non Aromatic

(2)  –Aromatic

(3)  –Aromatic

(4)  –Aromatic

44.[3]

	Zn	Cr	Fe	Co	Cu
Density/g cm^{-3}	7.1	7.19	7.8	8.7	8.9

45.[3] Clean water would have BOD value of less than 5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more.

46.[4]

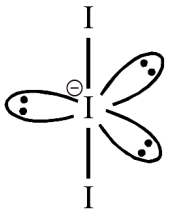
(i) In $VOSO_4$, 'V' is in +4 oxidation state. So it act as oxidising agent.

(ii) Cr_2O_3 is an amphoteric oxide.

(iii) In RuO_4, 'Ru' is in +8 oxidation state. So it act as oxidising agent.

(iv) Red colour of ruby is due to the presence of Cr^{+3} ions in Al_2O_3.

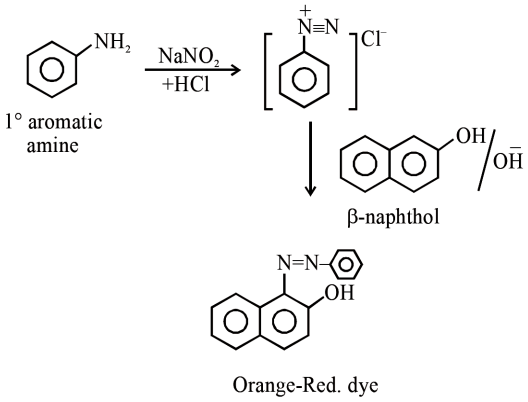
47.[4]



Shape : Linear, I–I–I Bond angle $\Rightarrow 180°$

48.[2] Most abundant gas in the troposphere is nitrogen.

49.[3]



50.[1] L.E. $\propto$ M.P.
L.E. : LiF > LiCl, MgO > NaCl

51.[3] Gabriel phthalimide synthesis is used to prepare 1° aliphatic/alicyclic amine in common. Hence amine which can synthesised by Gabriel phthalimide synthesis method is :
(A) $Me_2CH–CH_2–NH_2$ (B) $CH_3CH_2NH_2$
(C) $Ph–CH_2–NH_2$

52.[1] α-sulphur and β-sulphur are diamagnetic. S_2-form is paramagnetic.

53.[8] Combustion reaction :

$$C_xH_y(g)+\left(x+\frac{y}{4}\right)O_2(g)\rightarrow xCO_2(g)+\frac{y}{2}H_2O(\ell)$$

V — 6V

$$Vx = 4V$$

$$\Rightarrow \boxed{x=4}$$

Since : (I) $Vo_2 = 6 \times V_{C_xH_y}$

$$\Rightarrow V\left(x+\frac{y}{4}\right)=6V$$

$$\Rightarrow \boxed{\left(x+\frac{y}{4}\right)=6} \Rightarrow 4+\frac{y}{4}=6$$

$$\Rightarrow \boxed{y=8}$$

54.[5] Given Mass = 4.75g $\Rightarrow C_2H_2(g)$

$$\Rightarrow \text{Moles} = \frac{4.75}{26}\ \text{mol}$$

Temp. = 50 + 273 = 323 K

$$P=\frac{740}{760}\text{atm}$$

$$R = 0.0826\ \frac{\ell\text{atm}}{\text{molK}}$$

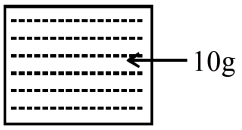
$$\Rightarrow V=\frac{nRT}{P}=\frac{4.75}{26}\times\frac{0.0826\times323}{\left(\frac{740}{760}\right)}$$

$$\Rightarrow V=\frac{96314.078}{19240}=5.0059\ell\simeq 5\ell$$

55.[1] Pure Solvent : C_6H_6 (ℓ)

Given : $T_f^\circ = 5.5°C$

$K_f = 5.12°C / m$



200 g C_6H_6

Solute is non dissociative

$\because \Delta T_f = k_f \times m$

$$\Rightarrow (T_f^\circ - T_f') = 5.12\times\frac{\left(\frac{10}{58}\right)}{\left(\frac{200}{1000}\right)\text{kg}}\text{mol}$$

$$5.5 - T_f' = \frac{5.12\times5\times10}{58}$$

$$\Rightarrow T_f' = 1.086°C \simeq 1°C$$

56.[3776]

Equation is-

$$MnO_4^- + H^\oplus + 5e^- \rightarrow Mn^{+2} + 4H_2O$$

Nernst equation :

$$E_{cell} = E_{cell}^\circ - \frac{0.059}{5}\log\frac{[Mn^{+2}]}{[MnO_4^-]}\left[\frac{1}{H^+}\right]^8$$

(I) Given $[H^\oplus] = 1M$

$$E_1 = E^0 - \frac{0.059}{5}\log\frac{[Mn^{+2}]}{[MnO_4^-]}$$

(II) Now : $[H^\oplus] = 10^{-4}$ M

$$E_2 = E^0 - \frac{0.059}{5}\log\frac{[Mn^{+2}]}{[MnO_4^-]}\times\frac{1}{(10^{-4})^8}$$

$$= E^0 - \frac{0.059}{5}\log\frac{[Mn^{+2}]}{[MnO_4^-]} \times \frac{0.059}{5}\log 10^{-32}$$

Therefore : $|E_1 - E_2| = \frac{0.059}{5} \times 32$

$= 0.3776\ V = 3776 \times 10^{-4}$

$x = 3776$

57.[141]

Given : $[K_{sp}]_{PbI_2} = 8 \times 10^{-9}$

To calculate: Solubility of PbI_2 in 0.1 M sol of $Pb(NO_3)_2$

(I) $Pb(NO_3)_2 \rightarrow Pb^{+2}_{(aq)} + 2NO_3^-(aq)$

$0.1\ M \quad - \quad 0.1M \quad 0.2M$

(II) $PbI_2(s) \rightleftharpoons Pb^{+2}(aq) + 2I^-(aq)$

$s \quad 2s$

$= s + 0.1$

$\simeq 0.1$

Now, : $K_{sp} = 8 \times 10^{-9} = [Pb^{+2}]\,[I^-]^2$

$\Rightarrow 8 \times 10^{-9} = 0.1 \times (2s)^2$

$\Rightarrow 8 \times 10^{-9} = 4s^2 \quad (s = \sqrt{2} \times 10^{-4})$

$\Rightarrow \boxed{s = 141 \times 10^{-6}\ M}$

$\Rightarrow x = 141$

58.[81] Given :

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow[t_{1/2}=\frac{10}{3}hr]{1\ order} \underset{Glucose}{C_6H_{12}O_6} + \underset{Fructose}{C_6H_{12}O_6}$$

$t = 0 \quad a = [A]_0$

$t = 9hr \quad a - x = [A]_t$

from I order kinetic : $\frac{k \times t}{2.303} = \log\frac{|A|_0}{|A|_t}$

$$\Rightarrow \frac{\ell n2 \times 9}{\frac{10}{3} \times 2.303} = \log\left(\frac{1}{f}\right)$$

$$\Rightarrow \frac{0.693 \times 9 \times 3}{23.03} = \log\left(\frac{1}{f}\right)$$

$$\Rightarrow \log\left(\frac{1}{f}\right) = 0.81246 = 81.24 \times 10^{-2}$$

$\Rightarrow x = 81$

59.[243]

$\boxed{M = 98}$ $\qquad$ $\boxed{M = 135}$

$$\underset{(Aniline)}{C_6H_5NH_2} \xrightarrow{90\%\ efficiency} \underset{(Acetanilide)}{C_6H_5\text{–}NH\text{–}\overset{\overset{O}{\|}}{C}\text{–}CH_3}$$

Given 1.86 g

$\Rightarrow$ 1 mol $C_6H_5NH_2$ give 1 mole $C_6H_5\ NHC\overset{\overset{O}{\|}}{C}CH_3$

$\therefore$ mole of $C_6H_5NH_2$ = moles of $C_6H_5\ NHC\overset{\overset{O}{\|}}{C}CH_3$

$$\Rightarrow \frac{1.86}{93} = \frac{W_{acetanilide}}{135}$$

$$\Rightarrow W_{acetanilide} = \frac{1.86 \times 135}{93}\,g = 2.70g$$

But efficiency of reaction is 90% only

$\therefore$ Mass of acetanilide produced = $2.70 \times \frac{90}{100}\,g$

$= 2.43\ g$

$= 243 \times 10^{-12}\ g$

$\Rightarrow x = 243$

60.[855]

$3HC \equiv CH_{(g)} \rightarrow C_6H_6(\ell) : \Delta G^0 = -RT\ell nk$

$\Delta G_f^0 - 2.04 \times 10^5 \frac{J}{mol} - 1.24 \times 10^5\ J/mol$

$\Rightarrow \Delta G^0 = \sum\left(\Delta G_f^0\right)_P - \sum\left(\Delta G_f^0\right)_R$

$\Rightarrow -RT\ell nk = 1 \times (-124 \times 10^5) - (-3 \times 2.04 \times 10^5)$

$\Rightarrow -2.303 \times R \times T \log k = 4.88 \times 10^5$

$$\Rightarrow \log k = -\frac{4.88 \times 10^5}{2.303 \times R \times T} = -\frac{488000}{5705.848} = 85.52$$

$= 855 \times 10^{-1}$

$\Rightarrow x = 855$

MATHEMATICS

61.[2] (A)

p	q	~q	p→q	~p	(~q ∧ (p → q))	
T	T	F	T	F	F	T
T	F	T	F	F	F	T
F	T	F	T	T	F	T
F	F	T	T	T	T	T

(B)

p	q	p ∨ q	~p	(p∨q)∧~p	
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

Both are tautologies

62.[1] P(9, 6, 9)

$$\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$$

$\Rightarrow Q = (20, b, -a, -9)$

$$\frac{\frac{20+a}{2}-3}{7} = \frac{\frac{b+6}{2}-2}{5} = \frac{-\frac{9}{2}-1}{-9}$$

$$\frac{14+9}{14} = \frac{b+2}{10} = \frac{a+2}{18}$$

$\Rightarrow$ a = –56 and b = –32

$\Rightarrow$ |a + b| = 88

63.[3] $\vec{r}.(\hat{i}+\hat{j}+\hat{k}) = 1$

$\vec{r}.(\hat{i}-2\hat{j}) = -2$

point (1, 0, 2)

Equation of plane

$\vec{r}.(\hat{i}+\hat{j}+\hat{k}) - 1 + \lambda\{r.(\hat{i}-2\hat{j}) + 2\} = 0$

$\vec{r}.\{\hat{i}(1+\lambda) + \hat{j}(1-2\lambda) + \hat{k}(1)\} - 1 + 2\lambda = 0$

point $\hat{i} + 0\hat{j} + 2\hat{k} = \vec{r}$

$\therefore (\hat{i}+2\hat{k})\{\hat{i}(1+\lambda) + \hat{j}(1-2\lambda) + \hat{k}(1)\} - 1 + 2\lambda = 0$

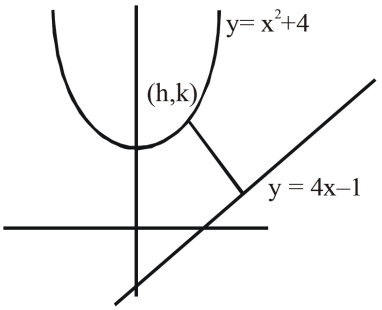
$1 + \lambda + 2 - 1 + 2\lambda = 0$

$$\lambda = -\frac{2}{3}$$

$$\therefore \vec{r}.\left[\hat{i}\left(\frac{1}{3}\right) + \hat{j}\left(\frac{7}{3}\right) + \hat{k}\right] = \frac{7}{3}$$

$\vec{r}.[\hat{i}+7\hat{j}+3\hat{k}] = 7$ Ans. 3

64.[4]



P : $y = x^2 + 4$

$k = h^2 + 4$

L : $y = 4x - 1$

$y - 4x + 1 = 0$

$$d = AB = \left|\frac{k-4h+1}{\sqrt{5}}\right| = \left|\frac{h^2-4-4h+1}{\sqrt{5}}\right|$$

$$\frac{d(d)}{dh} = \frac{2h-4}{\sqrt{5}} > 0$$

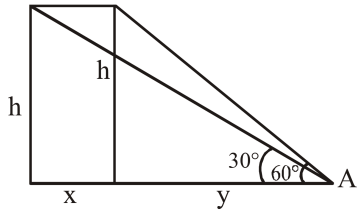
h = 2

$$\frac{d^2(d)}{dh^2} = \frac{2}{\sqrt{5}} > 0$$

$\therefore$ k = 4 + 4 = 8

$\therefore$ Point (2, 8)

65.[4]



$$\tan 60° = \frac{h}{y}$$

$$\sqrt{3} = \frac{h}{y} \Rightarrow h = \sqrt{3}y \quad .....(1)$$

$$\tan 30° = \frac{h}{x+y}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{x+y} \Rightarrow \sqrt{3}h = x + y \quad .....(2)$$

Speed 432 km/h $\Rightarrow \frac{432 \times 20}{60 \times 60} \Rightarrow \frac{12}{5}$ km

$$\sqrt{3}h = \frac{12}{5} + y$$

$$\sqrt{3}h - \frac{12}{5} = y$$

from (1)

$$h = \sqrt{3}\left[\sqrt{3}h - \frac{12}{5}\right]$$

$$h = 3h - \frac{12\sqrt{3}}{5}$$

$$h = \frac{6\sqrt{3}}{5} \text{ km}$$

$h = 1200\sqrt{3}$ m

66.[2] ${}^{n+1}C_2 + 2\,({}^2C_2 + {}^3C_2 + {}^4C_2 + ... + {}^nC_2)$

${}^{n+1}C_2 + 2\,({}^3C_3 + {}^3C_2 + {}^4C_2 + ... + {}^nC_2)$

{Use ${}^nC_{r+1} + {}^nC_r = {}^{n+1}C_r$}

$= {}^{n+1}C_2 + 2({}^4C_3 + {}^4C_2 + {}^5C_3 + ... + {}^nC_2)$

$= {}^{n+1}C_2 + 2({}^5C_3 + {}^5C_2 ..... + {}^nC_2)$

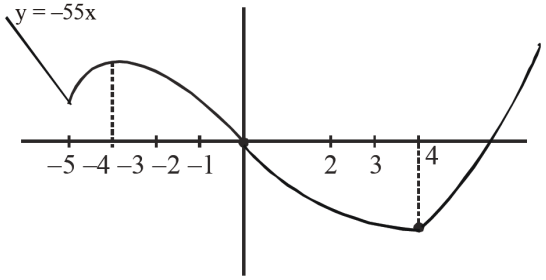
$= {}^{n+1}C_2 + 2({}^nC_3 + {}^nC_2)$

$= {}^{n+1}C_2 + 2\cdot {}^{n+1}C_3$

$$= \frac{(n+1)n}{2} + 2 \cdot \frac{(n+1)(n)(n-1)}{2.3}$$

$$= \frac{(n+1)n(2n+1)}{6}$$

67.[4]



$$f'(x)=\begin{cases}-55 & , \quad x<-5\\ 6(x-5)(x+4) & , \quad -5<x<4\\ 6(x-3)(x+2) & , \quad x>4\end{cases}$$

f(x) is increasing in

$x \in (-5,-4) \cup (4,\infty)$

68.[2] $f(x)f''(x)-(f'(x))^2=0$

$$\left[\frac{f''(x)}{f'(x)}=\frac{f'(x)}{f(x)}\right]$$

$\ln(f'(x)) = \ln f(x) + \ln c$

$f'(x) = cf(x)$

$$\frac{f'(x)}{f(x)}=c$$

$\ln f(x) = cx + k_1$

$f(x) = ke^{cx}$

$f(0) = 1 = k$

$f'(0) = c = 2$

$f(x) = e^{2x}$

$f(1) = e^2 \in (6, 9)$

69.[4] $m=-\frac{1}{\sqrt{3}}$, $c = 2$

(1) $c=a\sqrt{1+m^2}$

$$c=\sqrt{7}\frac{2}{\sqrt{3}} \quad \text{(incorrect)}$$

(2) $$c=\frac{a}{m}=\frac{\frac{1}{24\sqrt{3}}}{\frac{-1}{\sqrt{3}}}=-\frac{1}{24} \quad \text{(incorrect)}$$

(3) $c=\sqrt{a^2m^2-b^2}$

$$c=\sqrt{\frac{9}{2}\cdot\frac{1}{3}-\frac{1}{2}}=1 \quad \text{(incorrect)}$$

(4) $c=\sqrt{a^2m^2+b^2}$

$$c=\sqrt{9\cdot\frac{1}{3}+1}=2 \quad \text{(correct)}$$

70.[2] $$\int_1^3 \left(\left[(x-1)^2\right]-3\right)dx$$

$$=\int_1^3[x^2]-3\int_1^3 dx$$

$$\int_1^3 0dx+\int_1^{\sqrt{2}}1.dx+\int_{\sqrt{2}}^{\sqrt{3}}2.dx+\int_{\sqrt{3}}^{2}3.dx-6$$

$$=\sqrt{2}-1+2(\sqrt{3}-\sqrt{2})+3(2-\sqrt{3})-6$$

$$=-\sqrt{2}-\sqrt{3}-1$$

71.[1] Let $\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}=\theta$

$$\sin 4\theta=\frac{\sqrt{63}}{8}$$

$$\cos 4\theta=\frac{1}{8}$$

$$2\cos^2 2\theta-1=\frac{1}{8}$$

$$\cos^2 2\theta=\frac{9}{16}$$

$$\cos 2\theta=\frac{3}{4}$$

$$2\cos^2\theta-1=\frac{3}{4}$$

$$\cos^2\theta=\frac{7}{8}$$

$$\cos\theta=\frac{\sqrt{7}}{2\sqrt{2}}$$

$$\tan\theta=\frac{1}{\sqrt{7}}$$

72.[2] $\sim(\sim P \wedge (p \vee q))$

$p \vee (\sim p \wedge \sim q)$

$$\underbrace{(pv \sim p)}_{t} \wedge (pv \sim q)$$

$p \vee \sim q$

73.[3] $a + b + c = 2$ .....(1)

and $\left.\frac{dy}{dx}\right|_{(0,0)}=1$

$2ax + |_{(0,0)} = 1$

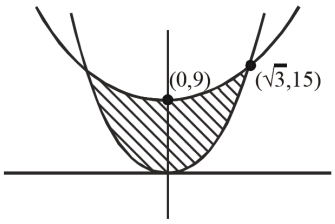
$b = 1$

Curve passes through origin

So, $c = 0$

and $a = 1$

74.[4]



Required area = $2\int_0^{\sqrt{3}}(2x^2+9-5x^2)dx$

$= 2\left[9x - x^3\right]_0^{\sqrt{3}}$

$= 2[9\sqrt{3} - 3\sqrt{3}] = 12\sqrt{3}$

75.[2] $\frac{dy}{dx} + \frac{y}{x} = bx^3$

I.F. $= e^{\int\frac{1}{x}dx} = x$

So, solution of D.E. is given by

$y.x = \int b.x^3, xdx + c$

$y = \frac{c}{x} + \frac{bx^4}{5}$

Passes through (1, 2)

$2 = c + \frac{b}{5}$ ..... (1)

$\int_1^2 f(x)dx = \frac{62}{5}$

$\left[c\ln x + \frac{bx^5}{25}\right]_1^2 = \frac{62}{5}$

$c\ln 2 + \frac{31b}{25} = \frac{62}{5}$ ..... (2)

By equation (1) & (2)

c = 0 and b = 10

76.[2] f ' (x) = f ' (2 –x)

f(x) = –f(2 – x) + c

put x = 0

f ' (0) = – f '(2) + c

c = f(0 + f(2) = 1 + e^2

so, f(x) + f(2 – x) = 1 + e^2

$I = \int_0^2 f(x)dx$

$I = \int_0^2 f(2-x)dx$

$2I = \int_0^2 (f(x) + f(2-x)dx$

$2I = (1+e^2)\int_0^2 dx$

$I = 1 + e^2$

77.[3] Let $A^T = A$ and $B^T = -B$

$C = A^2B^2 - B^2A^2$

$C^T = (A^2B^2)^T - (B^2A^2)^T$

$= (B^2)T(A^2)^T - (A^2)^T(B^2)^T$

$= B^2A^2 - A^2B^2$

$C^T = -C$

C is skew symmetric

So det(C) = 0

so system have infinite solutions

78.[4] $\frac{a+2+a}{3} = \frac{10}{3}$

a = 4 and $\frac{c+b+b}{3} = \frac{7}{3}$

c + 2b = 7

also 2b = a + c

2b – a + 2b = 7

$b = \frac{11}{4}$

now $4x^2 + \frac{11}{4}x + 1 = 0$ $\left\langle \begin{matrix}\alpha \\ \beta\end{matrix}\right.$

$\alpha^2 + \beta^2 - \alpha\beta = (\alpha+\beta)^2 - 3\alpha\beta$

$= \left(\frac{-11}{16}\right)^2 - 3\left(\frac{1}{4}\right)$

$= \frac{121}{256} - \frac{3}{4} = \frac{-71}{256}$

79.[4] $D = \begin{vmatrix}1 & -2 & 0\\ 1 & -1 & k\\ 0 & k & 4\end{vmatrix} = 4 - k^2$

so, A is correct and B,C, E are incorrect

If k = 2

$D_1 = \begin{vmatrix}1 & -2 & 0\\ -2 & -1 & 2\\ 6 & 2 & 4\end{vmatrix} = -48 \neq 0$

So no solution

D is correct

80.[3] Total subsets = $2^5 = 32$

Probability = $\frac{{}^5C_2 \times 3^3}{32 \times 32} = \frac{10 \times 27}{12^{10}} = \frac{135}{2^9}$

81.[BONUS]

Official Answer by NTA [12]

$$\sum_{i=0}^{k}\binom{10}{i}\binom{15}{k-i}+\sum_{i=0}^{k+1}\binom{12}{i}\binom{13}{k+1-i}$$

$^{25}C_k + {}^{25}C_{k+1}$

$^{26}C_{k+1}$

as ^nC_r is defined for all values of n as will as r so $^{26}C_{k+1}$ always exists

Now k is unbounded so maximum value is not defined

82.[1] $\frac{x-\lambda}{1}=\frac{y-\frac{1}{2}}{\frac{1}{2}}=\frac{z-0}{-\frac{1}{2}}$

$\frac{x-0}{1}=\frac{y+2\lambda}{1}=\frac{z-\lambda}{1}$

Shortest distance = $\frac{(a_2-a_1).(b_1\times b_2)}{|b_1\times b_2|}$

$$b_1\times b_2=\begin{vmatrix} i & j & k \\ 1 & \frac{1}{2} & -\frac{1}{2} \\ 1 & 1 & 1 \end{vmatrix}$$

$$=\hat{i}\left(\frac{1}{2}+\frac{1}{2}\right)-\hat{j}\left(1+\frac{1}{2}\right)+k\left(1-\frac{1}{2}\right)$$

$$\hat{i}-\frac{3}{2}\hat{j}+\frac{\hat{k}}{2}=\frac{2\hat{i}-3\hat{j}+\hat{k}}{2}$$

$$\frac{b_1\times b_2}{|b_1\times b_2|}=\frac{2\hat{i}-3\hat{j}+\hat{k}}{\sqrt{14}}$$

$$\frac{(a_2-a_1).(b_1\times b_2)}{|b_1\times b_2|}=\left(-\lambda\hat{i}+\left(-2\lambda+\frac{1}{2}\right)+\lambda\hat{k}\right)\left(\frac{2\hat{i}-3\hat{j}+\hat{k}}{\sqrt{14}}\right)$$

$$=\left|\frac{-2\lambda+6\lambda-\frac{3}{2}+\lambda}{\sqrt{14}}\right|=\frac{\sqrt{7}}{2\sqrt{2}}$$

$$\left|5\lambda-\frac{3}{2}\right|=\frac{7}{2}$$

$5\lambda=\frac{3}{2}\pm\frac{7}{2}$

$5\lambda = 5, -2$

$\lambda = 1, -\frac{2}{5}$

83.[2] $af(x)+\alpha f\left(\frac{1}{x}\right)=bx+\frac{\beta}{x}$ .... (1)

replace x by $\frac{1}{x}$

$af\left(\frac{1}{x}\right)+\alpha f(x)=\frac{b}{x}+\beta x$ .... (2)

(1) + (2)

$(a+\alpha)f(x)+(a+\alpha)f\left(\frac{1}{x}\right)=x(b+\beta)+(b+\beta)\frac{1}{x}$

$$\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}=\frac{b+\beta}{a+\alpha}=\frac{2}{1}=2$$

84.[56] Let point is (h, k)

So, $\sqrt{(h-5)^2+k^2}=3\sqrt{(h+5)^2+k^2}$

$8x^2 + 8y^2 + 100x + 200 = 0$

$x^2+y^2+\frac{25}{2}x+25=0$

$r^2=\frac{(25)^2}{4^2}-25$

$4r^2=\frac{25^2}{4}-100$

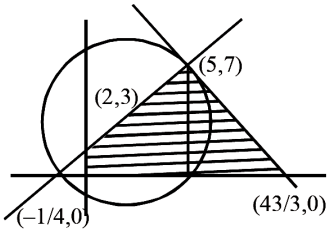
$4r^2 = 156.25 - 100$

$4r^2 = 56.25$

After round of $4r^2$ =56

85.[BONUS]

Official Answer by NTA [1225]



equation of normal

$4x - 3y + 1 = 0$

and equation of tangents

$3x + 4y - 43 = 0$

Area of triangle = $\frac{1}{2}\left(\frac{43}{3}+\frac{1}{4}\right)\times(7)$

$= \frac{1}{2}\left(\frac{172+3}{12}\right)\times 7$

$A = \frac{1225}{24}$

$24A = 1225$

* as positive x-axis is given in the question so question should be bonus.

86.[11] $\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$

$= \frac{9+k^2}{10} - \left(\frac{9+k}{10}\right)^2 < 10$

$90 + 10k^2 - 81 - k^2 - 18k < 1000$

$9k^2 - 18k - 991 < 0$

$k^2 - 2k < \frac{991}{9}$

$(k-1)^2 < \frac{1000}{9}$

$\frac{-10\sqrt{10}}{3} < k-1 < \frac{10\sqrt{10}}{3}$

$k < \frac{10\sqrt{10}}{3} + 1$

$k \le 11$

Maximum value of k is 11

87.[3] Let number are a, ar, ar^2, ar^3

$a\frac{(r^4-1)}{r-1} = \frac{65}{12}$ ...(1)

$\frac{1}{a}\left(\frac{\frac{1}{r4}-1}{\frac{1}{r}-1}\right) = \frac{65}{18}$

$\frac{1}{ar^3}\left(\frac{1-r^3}{1-r}\right) = \frac{65}{18}$ ....(2)

$\frac{(1)}{(2)} \Rightarrow a^2r^2 = \frac{3}{2}$

and $a^3.r^3 = 1$

$ar = 1$

$(ar)^2.r = \frac{3}{2}$

$r = \frac{3}{2}, a = \frac{2}{3}$

So, third term = $ar^2 = \frac{2}{3}\times\frac{9}{4}$

$\alpha = \frac{3}{2}$

$2\alpha = 3$

88.[31650]

If group C has one student then number of groups

$^{10}C_1[2^9 - 2] = 5100$

If group C has two students then number of groups

$^{10}C_2[2^8 - 2] = 11430$

If group C has three students then number of groups

$= {}^{10}C_3 \times [2^7 - 2] = 15120$

So total groups = 31650

89.[310] $K = \frac{1}{2^9}\left[\frac{\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{21}}{\left(\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2}}i\right)^{24}} + \frac{\left(\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)^{21}}{\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}i\right)^{24}}\right]$

$K = \frac{1}{512}\left[\frac{\left(e^{i\frac{2\pi}{3}}\right)^{21}}{\left(e^{-\frac{i\pi}{4}}\right)^{24}} + \frac{\left(e^{\frac{i\pi}{3}}\right)^{21}}{\left(e^{\frac{i\pi}{4}}\right)^{24}}\right]$

$K = \frac{1}{512}[e^{i(14\pi+6\pi)} + e^{i(7\pi-6\pi)}]$

$K = \frac{1}{512}[1+(-1)=0]$

$n = [|k|] = 0$

$\sum_{j=0}^{5}(j+5)^2 - \sum_{j=0}^{5}(j+5)$

$\sum_{j=0}^{5}(j^2+25+10j-j-5)$

$\sum_{j=0}^{5}(j^2+9j+20)$

$\sum_{j=0}^{5}j^2 + 9\sum_{j=0}^{5}j + 20\sum_{j=0}^{5}1$

$\frac{5\times6\times11}{6} + 9\left(\frac{5\times6}{2}\right) + 20\times6$

$= 55 + 135 + 120$

$= 310$

90.[2] Case-I

$x \le 5$

$(x+1)^2 - (x-5) = \frac{27}{4}$

$(x+1)^2 - (x+1) - \frac{3}{4} = 0$

$x+1 = \frac{3}{2}, -\frac{1}{2}$

$x = \frac{1}{2}, -\frac{3}{2}$

Case-II

$x > 5$

$(x+1) + (x-5) = \frac{27}{4}$

$(x+1)^2 + (x+1) - \frac{51}{4} = 0$

$x = \frac{-1 \pm \sqrt{52}}{2}$ (rejected as $x > 5$)

So, the equation have two real root